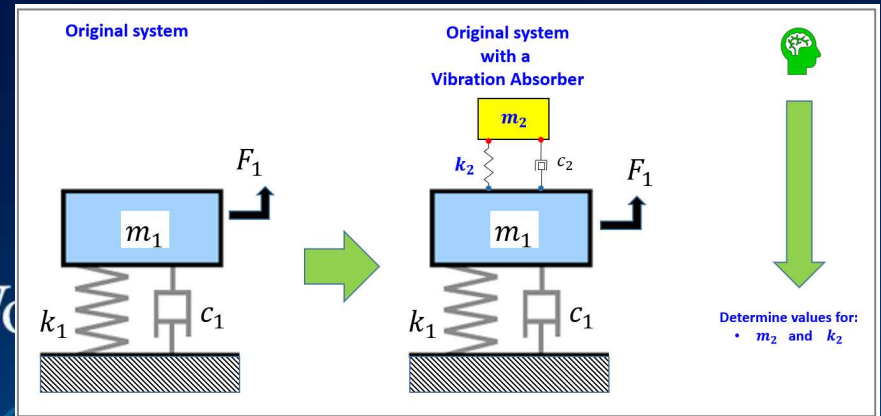
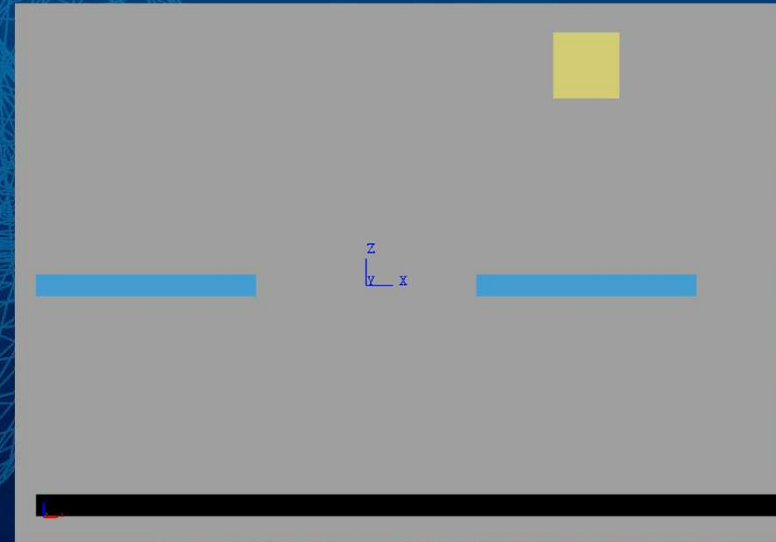


Today's Case Study



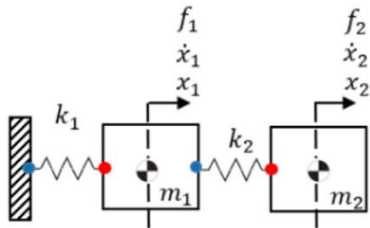
Design and simulate a Vibration absorber



Recall some Vibration absorber fundamentals - part 1 of 5

Appendix A: A Vibration Absorber

Consider the following undamped 2-dof system:



The equations of motion for this system are:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \cdot \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} + \begin{pmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & k_2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

which we can refer to as:

$$M\ddot{x} + Kx = f \quad \text{where} \quad X \in R^{n \times 1}$$

For a purely Sinusoidal excitation

$$f = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \cdot e^{j\omega t} = F \cdot e^{j\omega t}$$

We'll look for a **particular solution** of the form:

$$x = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \cdot e^{j\omega t} = X \cdot e^{j\omega t} \implies \ddot{x} = -\omega^2 \cdot X \cdot e^{j\omega t}$$

So we get:

$$M \cdot (-\omega^2 \cdot X \cdot e^{j\omega t}) + K \cdot X \cdot e^{j\omega t} = F \cdot e^{j\omega t}$$

$$M \cdot (-\omega^2 \cdot X) + K \cdot X = F$$

$$(-\omega^2 \cdot M + K) \cdot X = F$$

$$\left\{ \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \cdot \begin{bmatrix} -\omega^2 & 0 \\ 0 & -\omega^2 \end{bmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & k_2 \end{bmatrix} \right\} \cdot \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

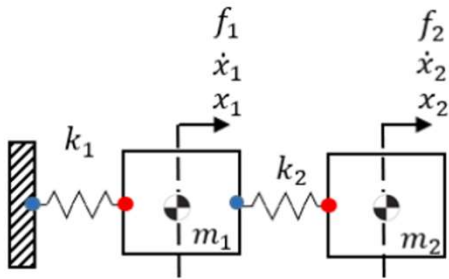
$$\begin{bmatrix} (-\omega^2 \cdot m_1 + k_1 + k_2) & -k_2 \\ -k_2 & (-\omega^2 \cdot m_2 + k_2) \end{bmatrix} \cdot \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \cdot \begin{bmatrix} (k_2 - \omega^2 \cdot m_2) & k_2 \\ k_2 & (k_1 + k_2 - \omega^2 \cdot m_1) \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

$$\begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \cdot \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix}$$

$$s = j \cdot \omega$$

Recall some Vibration absorber fundamentals - part 1 of 5

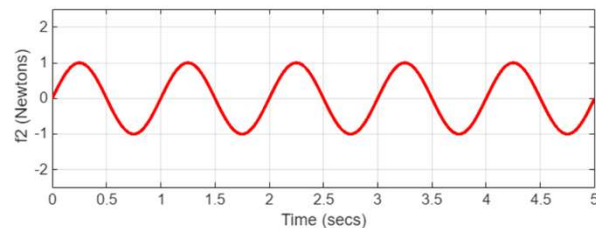
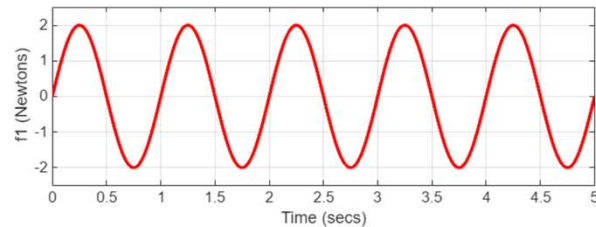


$$\begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \cdot \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix}$$

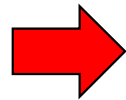
$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \cdot \begin{bmatrix} (k_2 - \omega^2 \cdot m_2) & k_2 \\ k_2 & (k_1 + k_2 - \omega^2 \cdot m_1) \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

$$\omega = 2\pi \left(\frac{\text{rad}}{\text{sec}} \right)$$

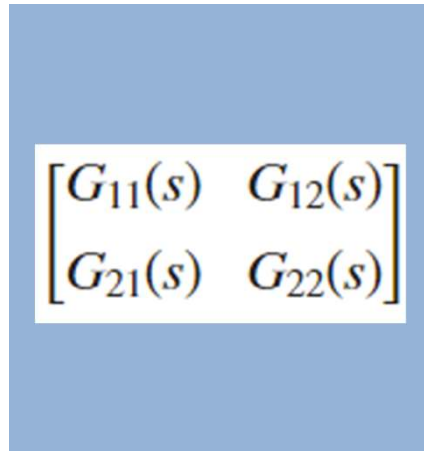
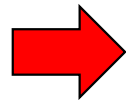
FORCE inputs: $f_1(t)$ and $f_2(t)$



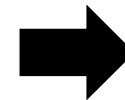
$f_1(t)$



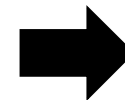
$f_2(t)$



$x_1(t)$

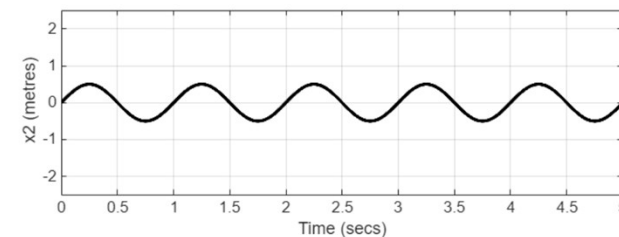
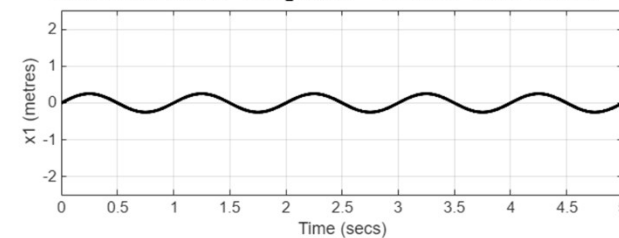


$x_2(t)$



$$\omega = 2\pi \left(\frac{\text{rad}}{\text{sec}} \right)$$

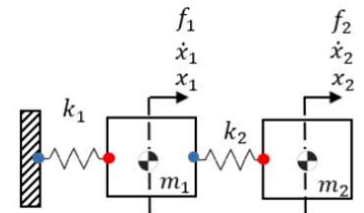
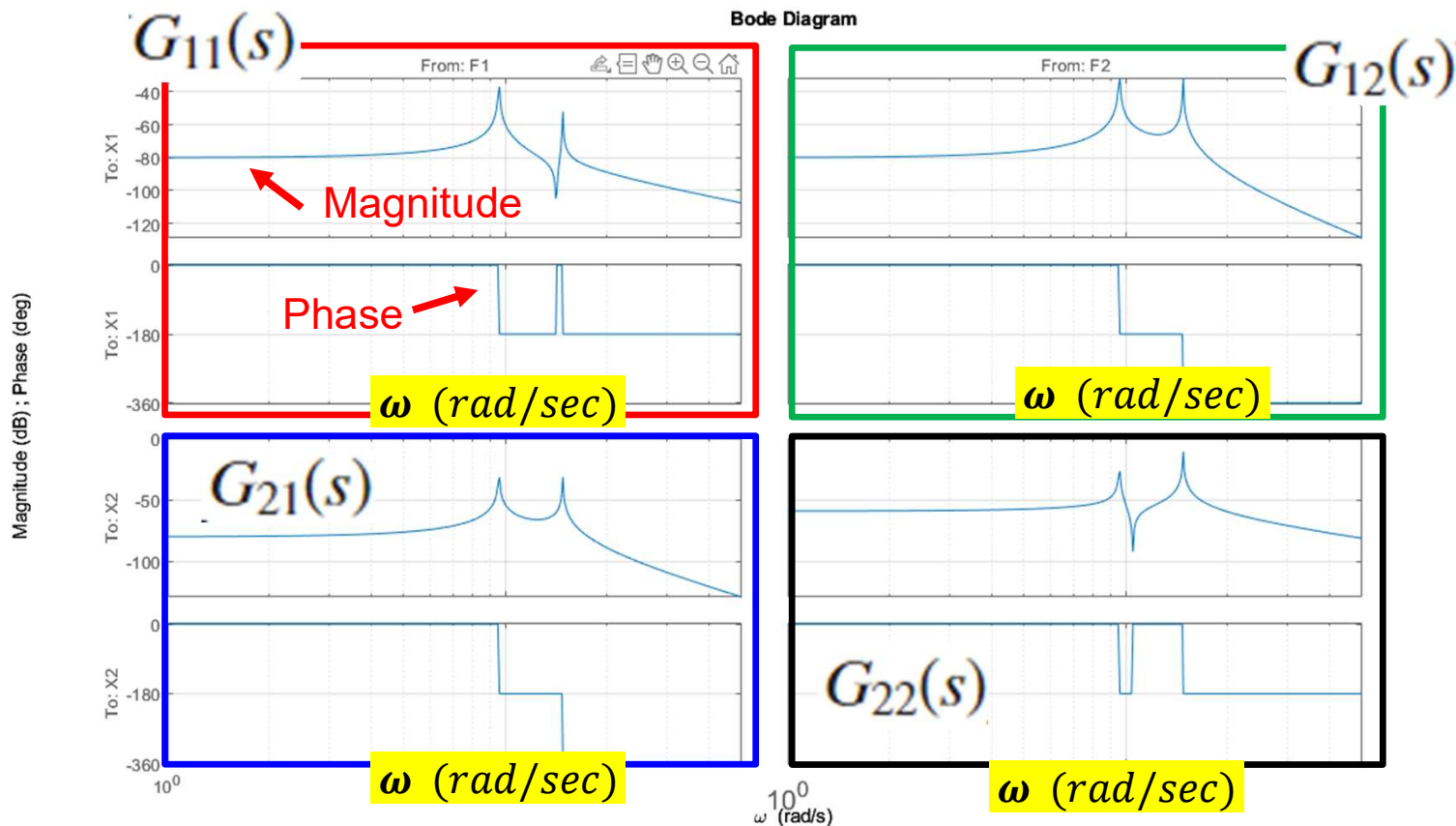
DISPLACEMENT outputs: $x_1(t)$ and $x_2(t)$



Recall some Vibration absorber fundamentals – part 2a of 5

$$\begin{bmatrix} X_1(s) \\ X_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix}$$

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \begin{bmatrix} (k_2 - \omega^2 \cdot m_2) & k_2 \\ k_2 & (k_1 + k_2 - \omega^2 \cdot m_1) \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

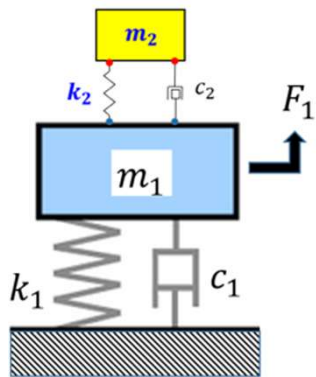
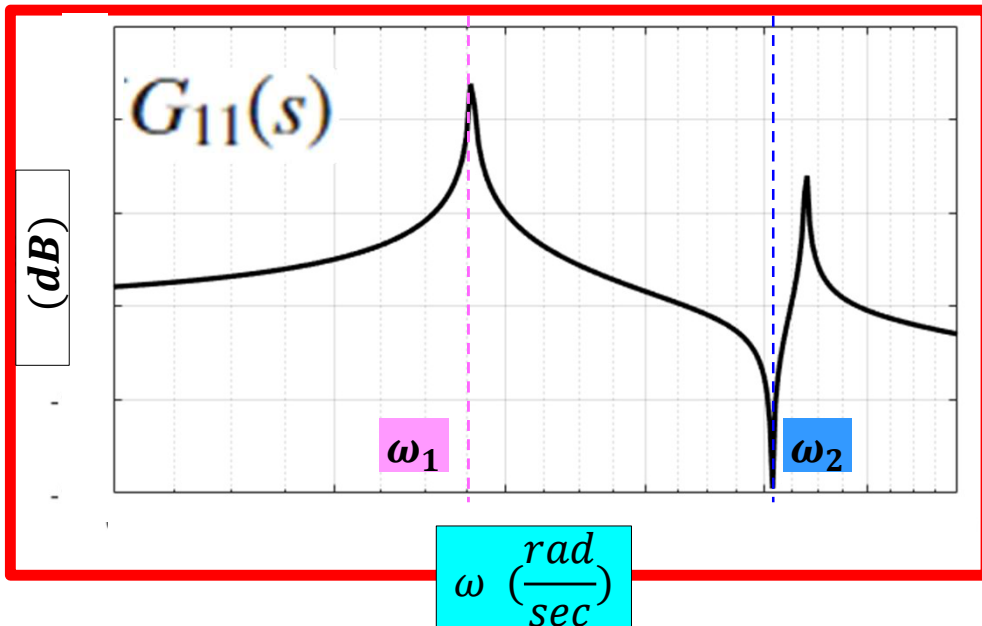


```
k1 = 1e4; % (N/m)
m1 = 1e2; % (kg)
k2 = 1e3; % (N/m)
m2 = 5; % (kg)
```

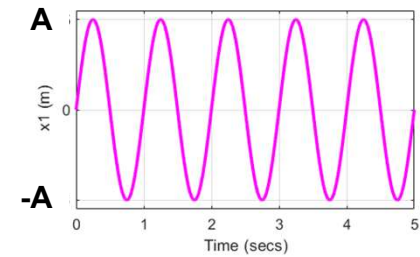
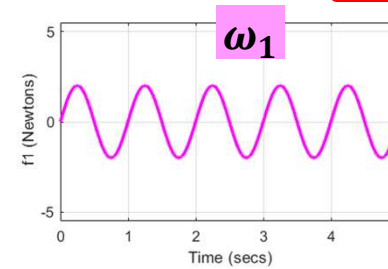


..and we've added a tiny ...
tiny bit of damping into this
system too ... which is why
we have PHASE shifts also

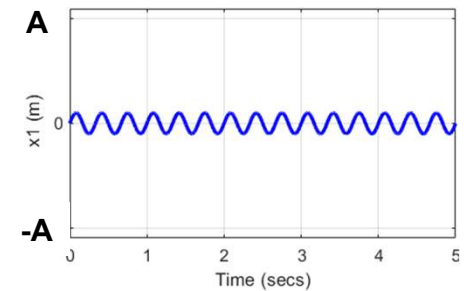
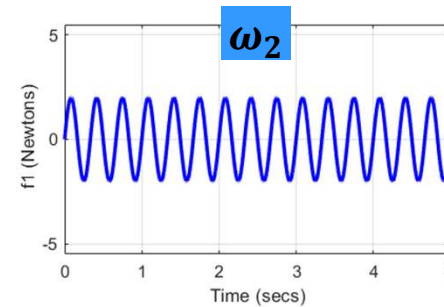
Recall some Vibration absorber fundamentals – part 2b of 5



$$f_1(t) \rightarrow \boxed{G_{11}(s)} \rightarrow x_1(t)$$



$$f_1(t) \rightarrow \boxed{G_{11}(s)} \rightarrow x_1(t)$$



Recall some Vibration absorber fundamentals – part 3 of 5

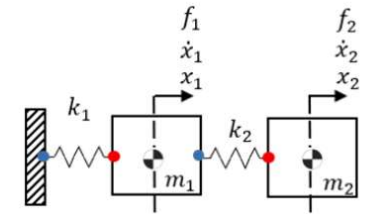
$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \begin{bmatrix} (k_2 - \omega^2 \cdot m_2) & k_2 \\ k_2 & (k_1 + k_2 - \omega^2 \cdot m_1) \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

For a **Vibration absorber**, we typically make $F_2 = 0$... which then leaves us with:

$$X_1(\omega) = \frac{(k_2 - \omega^2 \cdot m_2) \cdot F_1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \quad \text{and} \quad X_2(\omega) = \frac{k_2 \cdot F_1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2}$$

And we observe that:

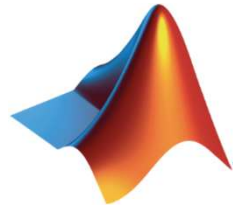
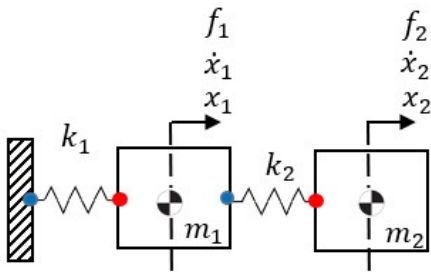
$$\begin{aligned} (k_2 = \omega^2 \cdot m_2) &\Rightarrow (0 = k_2 - \omega^2 \cdot m_2) \Rightarrow (X_1 = 0) \\ (k_2 = \omega^2 \cdot m_2) &\Rightarrow (0 = k_2 - \omega^2 \cdot m_2) \Rightarrow \left(X_2 = \frac{-F_1}{k_2} \right) \end{aligned}$$



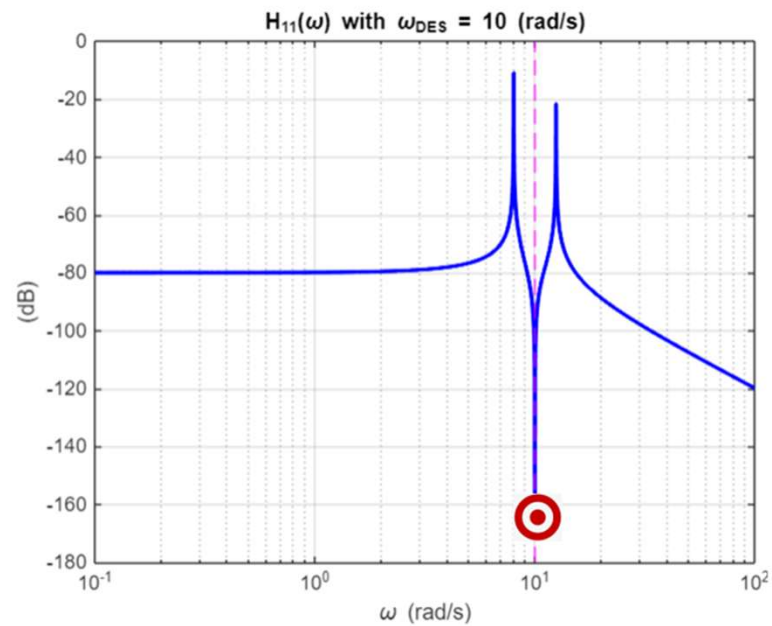
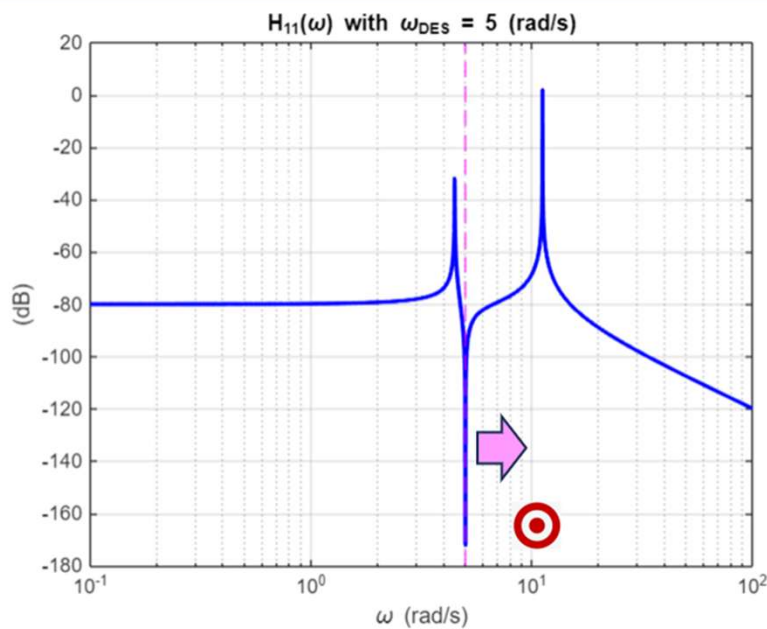
$$\begin{aligned} x &= \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \cdot e^{j\omega t} \\ f &= \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \cdot e^{j\omega t} \end{aligned}$$

- So ?
 - Is there ONE frequency that you want to place the ANTI resonance at ?
 - ω_{DES}
 - Pick a SPRING stiffness value
 - k_2
 - Pick a MASS value based on the formulae
 - $m_2 = \frac{k_2}{\omega_{DES}^2}$

Let's explore the placement of an ANTI resonance



[bh_FUNDAMENTALS_ABSORBER_explore.mlx](#)



Recall some Vibration absorber fundamentals – part 3 of 5

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \frac{1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \begin{bmatrix} (k_2 - \omega^2 \cdot m_2) & k_2 \\ k_2 & (k_1 + k_2 - \omega^2 \cdot m_1) \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

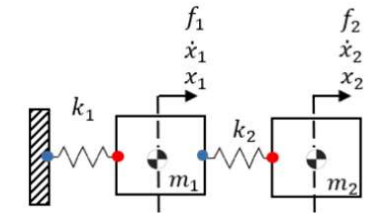
For a **Vibration absorber**, we typically make $F_2 = 0$... which then leaves us with:

$$X_1(\omega) = \frac{(k_2 - \omega^2 \cdot m_2) \cdot F_1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2} \quad \text{and} \quad X_2(\omega) = \frac{k_2 \cdot F_1}{(k_1 + k_2 - \omega^2 \cdot m_1) \cdot (k_2 - \omega^2 \cdot m_2) - k_2^2}$$

And we observe that:

$$\begin{aligned} (k_2 = \omega^2 \cdot m_2) &\implies (0 = k_2 - \omega^2 \cdot m_2) \implies (X_1 = 0) \\ (k_2 = \omega^2 \cdot m_2) &\implies (0 = k_2 - \omega^2 \cdot m_2) \implies \left(X_2 = \frac{-F_1}{k_2} \right) \end{aligned}$$

So we can design an absorber using the following procedure:



$$\begin{aligned} x &= \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \cdot e^{j\omega t} \\ f &= \begin{bmatrix} F_1 \\ F_2 \end{bmatrix} \cdot e^{j\omega t} \end{aligned}$$

Design Procedure

$$\omega_{\text{DES}} = \sqrt{\frac{k_1}{m_1}} \quad (\text{ie: at the undamped resonance of the MAIN machine})$$

1. Then we can choose k_2 such that: **OR maybe you're given k_2 !!**

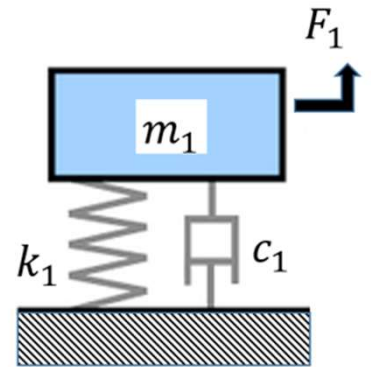
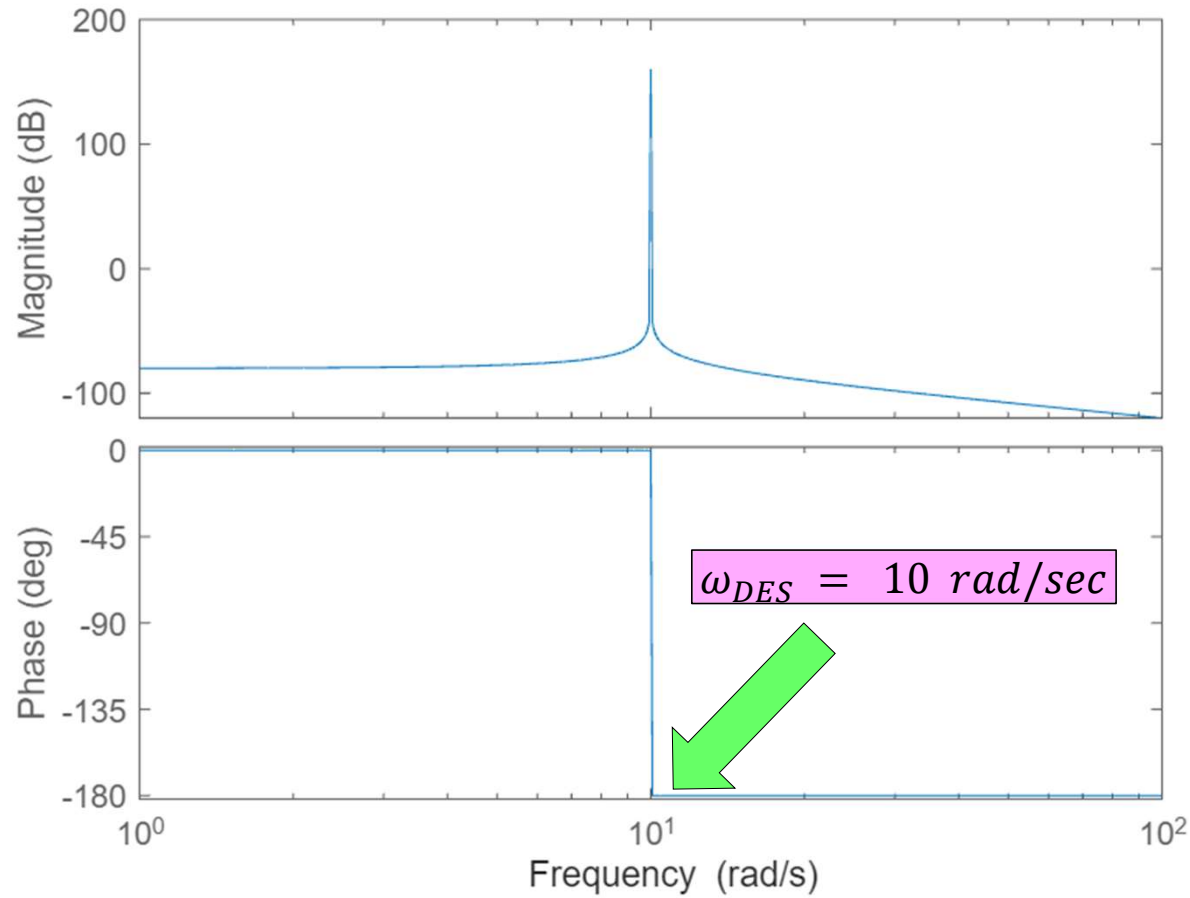
$$|X_2|_{\text{DES}} @ \omega_{\text{DES}} \implies k_2 = \left| \frac{F_1}{X_{2\text{DES}}} \right|$$

2. And we can choose m_2 such that:

$$X_1 = 0 @ \omega_{\text{DES}} \implies m_2 = \frac{k_2}{\omega_{\text{DES}}^2}$$

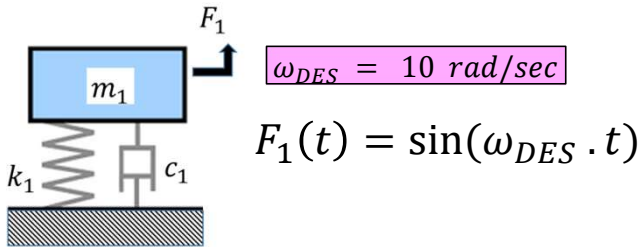
Recall some Vibration absorber fundamentals – part 4 of 5

Bode Diagram



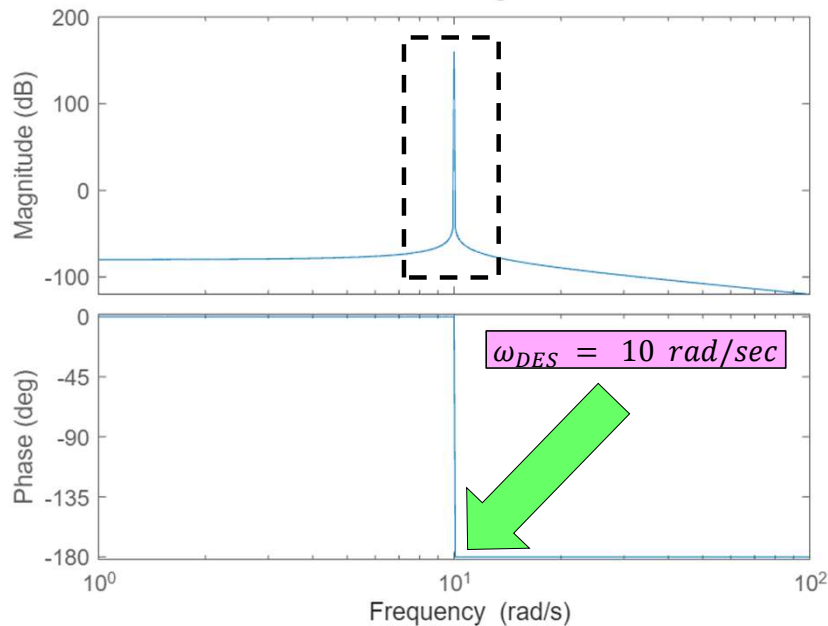
```
m1 = 100; % (kg)  
k1 = 1e4; % (N/m)
```

Recall some Vibration absorber fundamentals – part 5 of 5



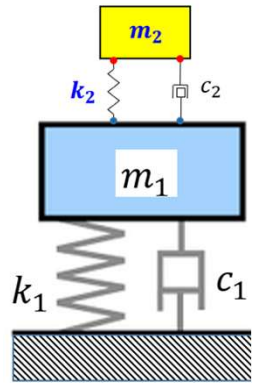
As a guide: $20 \cdot \log_{10}(10^6) = 120 \text{ dB}$

Bode Diagram



Design Procedure

- $\omega_{DES} = 10 \left(\frac{\text{rad}}{\text{sec}} \right)$
- $|X_2|_{DES} = 0.05 \text{ (m)}$ - a sensible value to start with
- $k_2 = \left| \frac{F_1}{X_2_{DES}} \right| = \frac{100}{0.05} = 2,000 \left(\frac{\text{N}}{\text{m}} \right)$
- $m_2 = \frac{k_2}{\omega_{DES}^2} = \frac{2000}{10^2} = 20 \text{ (kg)}$



As a guide: $20 \cdot \log_{10}(10^{-6}) = -120 \text{ dB}$

Bode Diagram

